



EAST WEST UNIVERSITY
Department of Computer Science and Engineering
B.Sc. in Computer Science and Engineering Program
Class Test-2, Summer 2026 Semester

Course: CSE 207- Data Structures, Section-1
Instructor: Dr. Maheen Islam, Associate Professor, CSE Department
Full Marks: 20
Time: 40 Minutes

Answer all questions. Write only the required functions.

Question 1

Write a recursive function that counts how many digits of a positive integer are greater than 5.

For example:

Input:

583764

Output:

Number of digits greater than 5: 4

Requirements:

- Use recursion.
- Do not use any loop.
- Write only the recursive function.

Question 2

Write a recursive function that calculates the sum of all leaf node values in a Binary Search Tree.

For example, consider the BST:

```
    50
   /  \
  30   70
 /  \   \
20 40  90
```

The leaf nodes are 20, 40, 90.

Output:

Sum of leaf nodes: 150

Requirements:

- Use recursion.
- Write only the function.
- Assume all values in the BST are distinct.